Shri Ramdeobaba College of Engineering & Management, Nagpur.
Department of Computer Science & Engineering
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Project Synopsis

Project Title: Smart Focus Distraction Control & Study Synchronization

Section: C

Group Number: 7

Group Member Details:

R.No	Name
42	Shruti Muley
49	Suyog Wasule
51	Tanaya Tapas
56	Vaibhavi Agrawal

Approved by

Prof. Barkha Dange
Prof. Samiha Khan

Problem Definition

In the digital age, the increasing use of the internet and smart devices has made information easily accessible, but it has also introduced significant distractions. Students often get diverted by social media, video platforms, and other entertainment websites, which negatively impacts their focus, productivity, and time management.

Although various website blockers and focus tools are available, most of them are designed for individual use. These tools lack features that promote accountability and motivation, allowing users to easily disable or bypass them when they lose focus. As a result, their effectiveness is limited and they fail to create long-term discipline.

Additionally, existing solutions do not support collaborative learning, where group study and peer accountability play an important role in maintaining consistency. This creates a gap between real-world study practices and digital tools.

Therefore, there is a need for a system that not only blocks distracting websites for a specific duration but also incorporates synchronized group study sessions and shared goals. Such a system can introduce accountability and motivation, helping users stay focused, improve productivity, and develop better digital habits.

Project Objectives

- **To implement network-level website blocking** using Chrome APIs, ensuring that distracting websites are restricted during focus sessions for improved concentration.
- **To enable real-time synchronized study sessions** among multiple users, allowing participants to stay connected and maintain discipline collectively.
- **To introduce a disruption mechanism**, where if one user leaves or breaks the session, it impacts all participants, thereby increasing accountability and encouraging commitment.
- **To track user activity and generate analytics**, such as focus time, distraction time, and productivity score, helping users understand and improve their study habits.

Proposed Methodology

The proposed system is implemented as a Chrome Extension based on Manifest V3 architecture, which consists of a popup interface for user interaction and a background service worker that handles the core functionalities of the system.

Initially, the user interacts with the extension through the popup interface, where they can define a list of distracting websites and set a timer for a focus session. These user inputs are processed and managed by the service worker, which acts as the central controller of the system.

For Solo Mode, the extension applies website blocking locally by using Chrome APIs. During an active session, access to selected websites is restricted, ensuring uninterrupted focus without requiring any external connectivity.

For Group Mode, the system integrates with Firebase Realtime Database to enable multi-user collaboration. Users can create or join study sessions using a unique room code. The session data, including timer state and participant activity, is stored and managed in the database.

To maintain synchronization, all connected clients continuously listen for updates from Firebase. Whenever there is a change in the session state, such as a timer update or a participant joining/leaving, the updated data is instantly reflected across all users.

An accountability mechanism is implemented where, if any participant exits the session before completion, the shared session state is updated in the database. This change is propagated to all connected users, resulting in termination of the session for everyone.

Additionally, the system tracks user activity such as focus time and distractions, which is later used to generate analytics and productivity insights. This helps users monitor their performance and improve their study habits.

Technology Stack

- Frontend Technologies: HTML, CSS, JavaScript
- Browser Platform: Chrome Extension APIs (Manifest V3)
- Backend: Firebase Realtime Database
- API Communication: REST APIs
- Local Storage: chrome.storage.local

Functional Specifications (Deliverables)

- A **Chrome Extension (Manifest V3 based)** for managing focus and controlling distractions.
- A **Solo Focus Mode** with a customizable timer and local website blocking functionality.
- A **network-level website blocking system** using Chrome APIs to restrict access to distracting sites.
- A **Group Study Module** with unique room codes for multiple users to join shared sessions.
- A **real-time synchronization mechanism** to maintain a consistent session state across all participants.
- An **accountability-based disruption feature**, where a session ends for all users if any participant exits early.
- An **Analytics Dashboard** that tracks user activity and displays productivity metrics such as focus time and distraction time.